

Predation rates based on dummy caterpillars

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2 Week 04 (November 23 - November 30)

Find 3 datasets from a domain specific data portal/repository.

<https://www.bexis.uni-jena.de/ddm/data/Showdata/31113>

<https://www.bexis.uni-jena.de/ddm/data/Showdata/24246>

<https://www.bexis.uni-jena.de/ddm/data/Showdata/25807>

Conduct an analysis manually on all 3 datasets.

Primary data

The sample dataset is available in Template(xlsm), Excel, CSV, TXT and TSV format.

The first dataset has 19 attributes namely Region, Start_date, End_date etc.

It has many missing data which are referred as NA.

It also has inconsistent number values for the same attribute.

Dummy_ID	Region	PlotID	Bexis_PlotID	Treatment	Subplot	Dummy	Arthropod_mark	Bird_mark	Mammal_mark	Gastropod_mark	Predation_total	Dummy_missing	Start_date	End_date	Max_veg_height	Average_veg_height
AEG01-EP_E1	A	AEG01	AEG1	EP	E	10	0	0	0	0	0	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-EP_E2	A	AEG01	AEG1	EP	E	20	0	0	0	0	0	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-EP_E3	A	AEG01	AEG1	EP	E	30	1	0	0	0	1	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-EP_E4	A	AEG01	AEG1	EP	E	40	1	0	0	0	1	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-EP_E5	A	AEG01	AEG1	EP	E	51	0	0	0	0	1	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-EP_E6	A	AEG01	AEG1	EP	E	61	0	0	0	0	1	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-EP_E7	A	AEG01	AEG1	EP	E	70	0	0	0	0	0	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-EP_E8	A	AEG01	AEG1	EP	E	80	0	0	0	0	0	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-EP_E9	A	AEG01	AEG1	EP	E	90	0	0	0	0	0	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-EP_E10	A	AEG01	AEG1	EP	E	100	0	0	0	0	0	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-EP_W1	A	AEG01	AEG1	EP	W	10	0	0	0	0	0	0	5/24/2021	5/26/2021	40.00	30.00
AEG01-EP_W2	A	AEG01	AEG1	EP	W	21	0	0	0	0	1	0	5/24/2021	5/26/2021	40.00	30.00
AEG01-EP_W3	A	AEG01	AEG1	EP	W	30	0	0	0	0	0	0	5/24/2021	5/26/2021	40.00	30.00
AEG01-EP_W4	A	AEG01	AEG1	EP	W	41	0	0	0	0	1	0	5/24/2021	5/26/2021	40.00	30.00
AEG01-EP_W5	A	AEG01	AEG1	EP	W	51	1	0	0	0	1	0	5/24/2021	5/26/2021	40.00	30.00
AEG01-EP_W6	A	AEG01	AEG1	EP	W	60	1	0	0	0	1	0	5/24/2021	5/26/2021	40.00	30.00
AEG01-EP_W7	A	AEG01	AEG1	EP	W	70	1	0	0	0	1	0	5/24/2021	5/26/2021	40.00	30.00
AEG01-EP_W8	A	AEG01	AEG1	EP	W	80	0	0	0	0	0	0	5/24/2021	5/26/2021	40.00	30.00
AEG01-EP_W9	A	AEG01	AEG1	EP	W	90	0	0	0	0	0	0	5/24/2021	5/26/2021	40.00	30.00
AEG01-EP_W10	A	AEG01	AEG1	EP	W	101	0	0	0	0	1	0	5/24/2021	5/26/2021	40.00	30.00
AEG01-RP_NE1	A	AEG01	AEG1	RP	NE	11	0	0	0	0	1	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-RP_NE2	A	AEG01	AEG1	RP	NE	20	0	0	0	0	0	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-RP_NE3	A	AEG01	AEG1	RP	NE	31	1	0	0	0	1	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-RP_NE4	A	AEG01	AEG1	RP	NE	40	0	0	0	0	0	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-RP_NE5	A	AEG01	AEG1	RP	NE	51	0	0	0	0	1	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-RP_NE6	A	AEG01	AEG1	RP	NE	60	1	0	0	0	1	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-RP_NE7	A	AEG01	AEG1	RP	NE	70	0	0	0	0	0	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-RP_NE8	A	AEG01	AEG1	RP	NE	80	0	0	0	0	0	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-RP_NE9	A	AEG01	AEG1	RP	NE	90	0	0	1	0	1	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-RP_NE10	A	AEG01	AEG1	RP	NE	100	0	0	0	0	0	0	5/24/2021	5/26/2021	50.00	30.00
AEG01-RP_SW1	A	AEG01	AEG1	RP	SW	10	1	0	0	0	1	0	5/24/2021	5/26/2021	45.00	30.00
AEG01-RP_SW2	A	AEG01	AEG1	RP	SW	20	0	0	0	0	0	0	5/24/2021	5/26/2021	45.00	30.00

The second dataset has 7 attributes namely Position, Side, Exploratorium etc. It has many missing data which are referred as NA.

It also has inconsistent number values for the same attribute.

The screenshot shows a spreadsheet titled '24246_2_data.xlsx - LibreOffice Calc'. The data is organized in a table with the following columns: A (Exploratorium), B (PlotID), C (Side), D (Position), E (PredationType), and F (PredationMark). The rows contain data for various plots (AEG01, AEG02) and sides (side1, side2, side3, side4). The 'PredationType' column lists 'Arthropod' and 'NA'. The 'PredationMark' column contains numerical values (0, 1) and 'NA'.

A	B	C	D	E	F
Exploratorium	PlotID	Side	Position	PredationType	PredationMark
ALB	AEG01	side2	AEG01-B-10	Arthropod	1
ALB	AEG01	side2	AEG01-B-11	Arthropod	1
ALB	AEG01	side2	AEG01-B-12	Arthropod	0
ALB	AEG01	side2	AEG01-B-9	Arthropod	NA
ALB	AEG01	side1	AEG01-B-8	Arthropod	0
ALB	AEG01	side1	AEG01-B-7	Arthropod	0
ALB	AEG01	side1	AEG01-B-5	Arthropod	0
ALB	AEG01	side1	AEG01-B-6	Arthropod	0
ALB	AEG01	side3	AEG01-B-13	Arthropod	0
ALB	AEG01	side3	AEG01-B-15	Arthropod	0
ALB	AEG01	side3	AEG01-B-16	Arthropod	1
ALB	AEG01	side3	AEG01-B-14	Arthropod	0
ALB	AEG01	side4	AEG01-B-1	Arthropod	0
ALB	AEG01	side4	AEG01-B-3	Arthropod	0
ALB	AEG01	side4	AEG01-B-4	Arthropod	0
ALB	AEG01	side4	AEG01-B-2	Arthropod	0
ALB	AEG02	side2	AEG02-B-7	Arthropod	1
ALB	AEG02	side2	AEG02-B-8	Arthropod	1
ALB	AEG02	side2	AEG02-B-5	Arthropod	0
ALB	AEG02	side2	AEG02-B-6	Arthropod	NA
ALB	AEG02	side1	AEG02-B-3	Arthropod	0
ALB	AEG02	side1	AEG02-B-1	Arthropod	1
ALB	AEG02	side1	AEG02-B-4	Arthropod	1
ALB	AEG02	side1	AEG02-B-2	Arthropod	0
ALB	AEG02	side3	AEG02-B-10	Arthropod	1
ALB	AEG02	side3	AEG02-B-11	Arthropod	0
ALB	AEG02	side3	AEG02-B-12	Arthropod	1
ALB	AEG02	side3	AEG02-B-9	Arthropod	0
ALB	AEG02	side4	AEG02-B-14	Arthropod	1
ALB	AEG02	side4	AEG02-B-15	Arthropod	NA
ALB	AEG02	side4	AEG02-B-13	Arthropod	0
ALB	AEG02	side4	AEG02-B-16	Arthropod	0

The first dataset has 13 attributes namely Region, Subplot, Nr_dummies etc.

It has many missing data which are referred as NA.

It also has inconsistent number values for the same attribute.

The screenshot shows a spreadsheet titled '25807_3_data.xlsx - LibreOffice Calc'. The data is organized in a table with the following columns: A (PlotID), B (Region), C (Subplot), D (Date_start), E (Time_start), F (Date_end), G (Time_end), H (Nr_dummies), I (Arthropod_prop), J (Bird_prop), K (Mammal_prop), L (Gastropod_prop), M (Predation_prop), N (Predation_prop), O (Predation_prop), P (Predation_prop), Q (Predation_prop), R (Predation_prop), S (Predation_prop), T (Predation_prop), U (Predation_prop). The rows contain data for various plots (AEG01, AEG02, AEG10, AEG11, AEG12) and regions (A, B, C, D, E). The 'Nr_dummies' column contains numerical values (8, 9, 10). The 'Arthropod_prop' column contains numerical values (0.00, 0.11, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, 1.00). The 'Bird_prop' column contains numerical values (0.00, 0.11, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, 1.00). The 'Mammal_prop' column contains numerical values (0.00, 0.11, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, 1.00). The 'Gastropod_prop' column contains numerical values (0.00, 0.11, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, 1.00). The 'Predation_prop' column contains numerical values (0.00, 0.11, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, 1.00).

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
PlotID	Region	Subplot	Date_start	Time_start	Date_end	Time_end	Nr_dummies	Arthropod_prop	Bird_prop	Mammal_prop	Gastropod_prop	Predation_prop	Predation_prop	Predation_prop	Predation_prop	Predation_prop	Predation_prop	Predation_prop	Predation_prop	Predation_prop
AEGW6	A	B	6/9/2017	12/30/1899	6/11/2017	12/30/1899	10	0.00	0.00	0.70	0.40	0.80								
AEGW5	A	E	6/26/2017	12/30/1899	12/30/1899	12/30/1899	10	0.00	0.00	0.60	0.50	0.70								
AEG28	A	B	6/21/2017	12/30/1899	12/30/1899	12/30/1899	10	0.00	0.00	0.00	0.00	0.00								
AEG13	A	E	6/8/2017	12/30/1899	6/10/2017	12/30/1899	9	0.11	0.00	0.22	0.11	0.44								
AEG28	A	C	6/21/2017	12/30/1899	12/30/1899	12/30/1899	9	0.00	0.00	0.00	0.00	0.00								
AEG28	A	D	6/21/2017	12/30/1899	12/30/1899	12/30/1899	10	0.00	0.00	0.00	0.00	0.00								
AEG28	A	E	6/21/2017	12/30/1899	12/30/1899	12/30/1899	9	0.00	0.00	0.00	0.00	0.00								
AEG26	A	B	6/13/2017	12/30/1899	6/15/2017	12/30/1899	4	0.00	0.00	0.00	0.00	0.00								
AEG13	A	D	6/8/2017	12/30/1899	6/10/2017	12/30/1899	10	0.30	0.00	0.20	0.00	0.50								
AEG13	A	C	6/8/2017	12/30/1899	6/10/2017	12/30/1899	10	0.10	0.00	0.00	0.10	0.20								
AEG13	A	B	6/8/2017	12/30/1899	6/10/2017	12/30/1899	10	0.30	0.00	0.30	0.00	0.50								
AEG13	A	A	6/8/2017	12/30/1899	6/10/2017	12/30/1899	10	0.10	0.00	0.30	0.00	0.40								
AEG12	A	E	6/8/2017	12/30/1899	6/10/2017	12/30/1899	10	0.50	0.00	0.20	0.00	0.60								
AEG1	A	A	6/8/2017	12/30/1899	6/10/2017	12/30/1899	10	0.80	0.00	0.30	0.20	0.90								
AEG1	A	B	6/8/2017	12/30/1899	6/10/2017	12/30/1899	10	0.90	0.00	0.00	0.10	0.90								
AEG1	A	C	6/8/2017	12/30/1899	6/10/2017	12/30/1899	10	0.70	0.00	0.20	0.00	0.80								
AEG1	A	D	6/8/2017	12/30/1899	6/10/2017	12/30/1899	10	0.60	0.00	0.30	0.00	0.80								
AEG1	A	E	6/8/2017	12/30/1899	6/10/2017	12/30/1899	10	0.70	0.00	0.00	0.00	0.70								
AEG10	A	A	6/20/2017	12/30/1899	6/22/2017	12/30/1899	10	0.00	0.00	0.00	0.00	0.00								
AEG10	A	B	6/20/2017	12/30/1899	6/22/2017	12/30/1899	8	0.00	0.00	0.00	0.00	0.00								
AEG10	A	C	6/20/2017	12/30/1899	6/22/2017	12/30/1899	9	0.00	0.00	0.00	0.00	0.00								
AEG10	A	D	6/20/2017	12/30/1899	6/22/2017	12/30/1899	10	0.00	0.00	0.00	0.00	0.00								
AEG10	A	E	6/20/2017	12/30/1899	6/22/2017	12/30/1899	10	0.00	0.00	0.10	0.00	0.10								
AEG11	A	A	6/27/2017	12/30/1899	6/29/2017	12/30/1899	9	0.11	0.00	0.00	0.00	0.11								
AEG11	A	B	6/27/2017	12/30/1899	6/29/2017	12/30/1899	10	0.00	0.00	0.00	0.00	0.00								
AEG11	A	C	6/27/2017	12/30/1899	6/29/2017	12/30/1899	10	0.00	0.00	0.00	0.00	0.00								
AEG11	A	D	6/27/2017	12/30/1899	6/29/2017	12/30/1899	10	0.00	0.00	0.00	0.10	0.10								
AEG11	A	E	6/27/2017	12/30/1899	6/29/2017	12/30/1899	10	0.00	0.00	0.00	0.00	0.00								
AEG12	A	A	6/8/2017	12/30/1899	6/10/2017	12/30/1899	9	0.33	0.00	0.11	0.11	0.44								
AEG12	A	B	6/8/2017	12/30/1899	6/10/2017	12/30/1899	10	0.40	0.00	0.20	0.00	0.50								
AEG12	A	C	6/8/2017	12/30/1899	6/10/2017	12/30/1899	10	0.40	0.00	0.00	0.00	0.40								
AEG12	A	D	6/8/2017	12/30/1899	6/10/2017	12/30/1899	10	0.60	0.00	0.10	0.00	0.60								
AEG12	A	E	6/8/2017	12/30/1899	6/10/2017	12/30/1899	10	0.50	0.00	0.00	0.00	0.50								

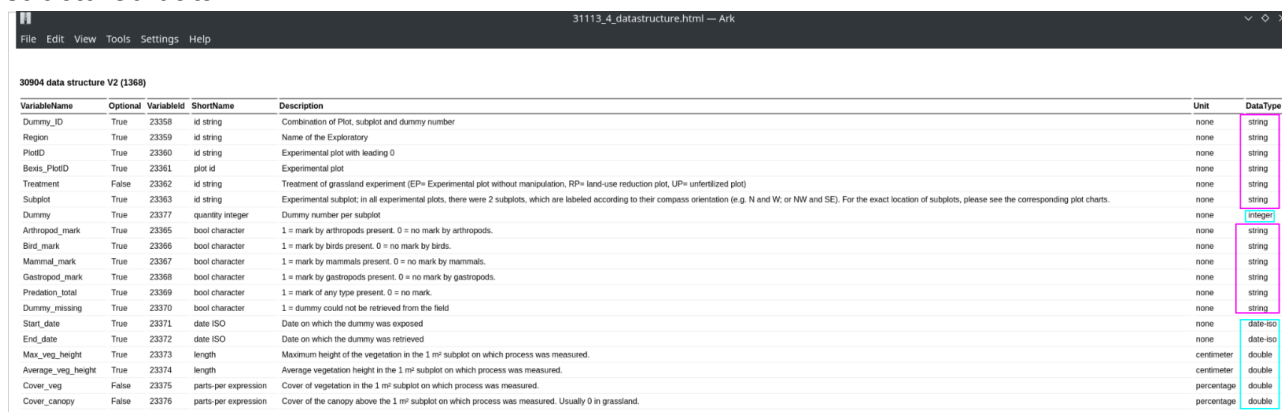
Metadata

The metadata of datasets is available in HTML and XML format

These datasets have many fields of metadata such as title, abstract, Project name, creator, methods, creation date, modification date, license etc.

Type of Data

These datasets are mixture of qualitative and quantitative data and have structured data.



VariableName	Optional	VariableID	ShortName	Description	Unit	DataType
Dummy_ID	True	23358	id string	Combination of Plot, subplot and dummy number	none	string
Region	True	23359	id string	Name of the Exploratory	none	string
PlotID	True	23360	id string	Experimental plot with leading 0	none	string
Beats_PlotID	True	23361	plot id	Experimental plot	none	string
Treatment	False	23362	id string	Treatment of grassland experiment (EP= Experimental plot without manipulation, RP= land-use reduction plot, UP= unfertilized plot)	none	string
Subplot	True	23363	id string	Experimental subplot, in all experimental plots, there were 2 subplots, which are labeled according to their compass orientation (e.g. N and W, or NW and SE). For the exact location of subplots, please see the corresponding plot charts.	none	string
Dummy	True	23377	quantity integer	Dummy number per subplot	none	integer
Arthropod_mark	True	23365	bool character	1 = mark by arthropods present, 0 = no mark by arthropods.	none	string
Bird_mark	True	23366	bool character	1 = mark by birds present, 0 = no mark by birds.	none	string
Mammal_mark	True	23367	bool character	1 = mark by mammals present, 0 = no mark by mammals.	none	string
Gastropod_mark	True	23368	bool character	1 = mark by gastropods present, 0 = no mark by gastropods.	none	string
Predation_total	True	23369	bool character	1 = mark of any type present, 0 = no mark.	none	string
Dummy_missing	True	23370	bool character	1 = dummy could not be retrieved from the field	none	string
Start_date	True	23371	date ISO	Date on which the dummy was exposed	none	date-iso
End_date	True	23372	date ISO	Date on which the dummy was retrieved	none	date-iso
Max_veg_height	True	23373	length	Maximum height of the vegetation in the 1 m ² subplot on which process was measured.	centimeter	double
Average_veg_height	True	23374	length	Average vegetation height in the 1 m ² subplot on which process was measured.	centimeter	double
Cover_veg	False	23375	parts-per expression	Cover of vegetation in the 1 m ² subplot on which process was measured.	percentage	double
Cover_canopy	False	23376	parts-per expression	Cover of the canopy above the 1 m ² subplot on which process was measured. Usually 0 in grassland.	percentage	double

We answered the following questions for all of our datasets.

How the data is created?

First dataset: <https://www.bexis.uni-jena.de/ddm/data/Showdata/31113>

Manual collection with equipments such as insect needles, cardboard boxes and stereo microscope.

Second dataset: <https://www.bexis.uni-jena.de/ddm/data/Showdata/24246>

Manual collection with equipments such as petri dishes, filter paper, artificial diet (based on whey protein, caseinate, egg powder, sucrose, and water, gelled with agar), plastic cups filled with a solution of water, salt and soap.

Third dataset: <https://www.bexis.uni-jena.de/ddm/data/Showdata/25807>

Manual collection with equipments such as insect needles, cardboard boxes and stereo microscope.

How is it documented?

All of our datasets are available in Template(xlsm), Excel, CSV, TXT and TSV format.

Who will be able to access it?

All of our datasets have CC BY 4.0 license which means the data is OpenData.

Where the data is stored?

The data is stored under <https://www.bexis.uni-jena.de/ddm/data/Showdata/31113> , <https://www.bexis.uni-jena.de/ddm/data/Showdata/24246> and <https://www.bexis.uni-jena.de/ddm/data/Showdata/25807>

Who will back it up?

No information available on this matter in any of our datasets.

Whether and how it will be shared and preserved?

The data is shared on a data portal which is preserved for all of our datasets.

Cost and resources?

All of our datasets are funded by German Research Foundation <https://ror.org/018mejw64>

Duties and responsibilities?

First dataset: <https://www.bexis.uni-jena.de/ddm/data/Showdata/31113>

The data contain the presence/absence of attack marks separated by arthropods, small mammals, birds and snails. As potential covariates, vegetation cover, vegetation height and canopy cover are given.

Second dataset: <https://www.bexis.uni-jena.de/ddm/data/Showdata/24246>

This dataset explores land use impact on arthropods in German grasslands. It measures predation attempts, arthropod abundance, and ant activity across different land-use intensities.

Third dataset: <https://www.bexis.uni-jena.de/ddm/data/Showdata/25807>

This study investigates predation on dummy caterpillars, comparing forests and semi-natural grasslands in three regions. It examines correlations between process rates, and the impact of climate and land-use intensity at short and medium-term scales.

Contributions

Shirin Shoghli

Discussing, searching and selecting datasets in the animal domain, specifying primary data and metadata, analyzing data type and answering questions based on DMP perspective about both primary data and metadata.

Shahrzad Torabi

Discussing, searching and selecting datasets in the animal domain, specifying primary data and metadata, analyzing data type and answering questions based on DMP perspective about both primary data and metadata.

Sanaz Bayat

Discussing, searching and selecting datasets in the animal domain, specifying primary data and metadata, analyzing data type and answering questions based on DMP perspective about both primary data and metadata.

Maryam Gheibi

Discussing, searching and selecting datasets in the animal domain, specifying primary data and metadata, analyzing data type and answering questions based on DMP perspective about both primary data and metadata.

Mozhdeh Ramezani Dastjerdi

Discussing, searching and selecting datasets in the animal domain, specifying primary data and metadata, analyzing data type and answering questions based on DMP perspective about both primary data and metadata.